

2.4 RENEWABLE ENERGY AND ENERGY EFFICIENCY

Waverley Council has set an ambitious target to reduce community greenhouse emissions to net zero by 2035. In order to meet this reduction target, all new homes are required to have future capacity to be an all-electric building, powered only by renewable energy.

To achieve net zero by 2035, installing natural gas appliances in new developments is not recommended.

Fluorescent and compact fluorescent lamps contain small amounts of mercury, a highly toxic agent which bioaccumulates in the environment. Recycling rates of fluorescent lamps are as low as 2% (Environment Victoria, 2022). For this reason, Waverley Council supports energy efficient alternatives to fluorescent lamps, such as Light Emitting Diodes (LEDs).

Energy efficiency measures for new residential developments are stipulated under the State Environmental Planning Policy (Building Sustainable Index: BASIX) 2004. Commercial energy efficiency measures are stipulated under the National Construction Code Section J.

Objectives

- (a) To enable all development to contribute to net zero greenhouse emissions by 2035.
- (b) To reduce the energy demand of all developments.
- (c) To ensure a building can be 100% powered by renewable energy.
- (d) To encourage the installation and use of renewable energy technologies to reduce greenhouse emissions and peak demand.
- (e) To ensure development takes into consideration neighbouring solar technologies in the design of the building.

Controls

Solar photovoltaic system and battery

- (a) The installation of photovoltaic panels with battery storage is strongly encouraged in all developments.
- (b) Developments proposing to install photovoltaic panels in heritage conservation areas must refer to the Heritage on Solar guidelines on Council's website.

Domestic hot water

- (c) An electric hot water system is strongly encouraged in all developments. Recommended systems include:
 - Electric heat pump (most efficient)
 - Solar thermal with electric boost (most efficient)
 - Electric storage